

Assessment of autonomic function tests and predictive features for cardiac autonomic neuropathy in T2-DM

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Introduction

Worldwide, approximately 14% of the population is affected by diabetes mellites, corresponding to 800 million individuals [1]. Diabetes mellites is a chronic metabolic health condition characterized by the inability to maintain glycemic control [2]. This increasing prevalence of people with diabetes is an incentive for improving treatments [3].

In people with diabetes, the primary focus of treatment is optimized glycemic control. A well-regulated glycemic status is obtained if the glucose concentration is maintained between 4 – 10 mmol/l, 70% of the time [4]. Continuous Glucose Management (CGM) contributes to the awareness of blood glucose levels [3]. Well-regulated diabetes with appropriate glycemic control can reduce the incidence of comorbidities [5].

When people with diabetes experience fluctuating or sustained high glucose concentrations, this increases the probability for developing comorbidities [6]. A prevalent comorbidity of diabetes is cardiac autonomic neuropathy (CAN) [7]. In this diabetes induced neuropathy, autonomic regulation is impaired by a pathological process causing systemic neuronal inflammation and dysfunction [8]. This neuronal inflammation is able to cause harmful cardiac dysfunction [7,8]. CAN and other comorbidities lead to a 3.7

fold higher occurrence of cardiac death in T1DM compared to healthy controls [9].

To identify CAN the Cardiac Autonomic Reflex Tests (CART) were introduced [10]. These CART tests are hard to implement. The testing battery requires the patient to undergo a 20-minute lasting intense testing protocol. Therefore, the implementation of CART test for CAN identification requires significantly large time- and contributions. Therefore, halting the implementation of CART testing on the large volume of people with diabetes in clinical practice.

The hypothesis is that if patient history, blood tests and autonomic tests are analyzed, we can identify CAN with machine learning (ML) models. Therefore, reducing the effort required for clinical implementation of CAN-screening. The validation of low effort predictors against high burden CART tests could facilitate the implementation of CAN identification in medical practice.

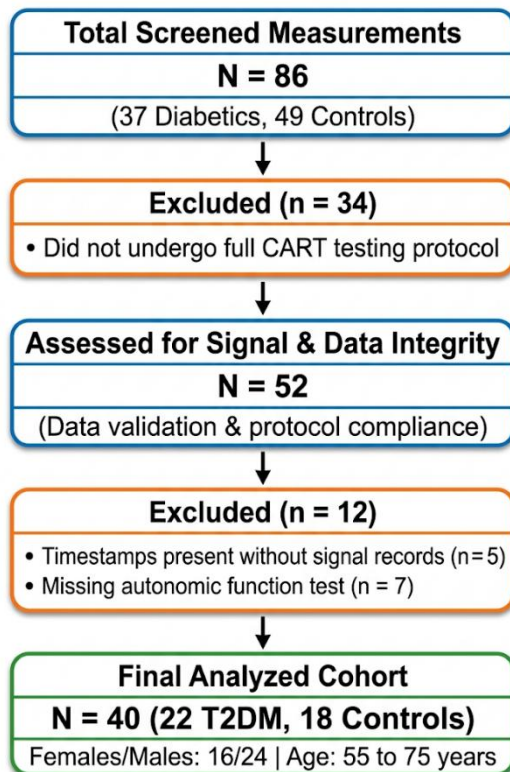
Methods

Study design and clinical rationale

This is a secondary subanalysis of cross-sectional analysis (n=40), From of n=86, N=46 patients had to be excluded. The data originated from open-source Physionet dataset named: “Cerebral Vaso regulation in Diabetes”. This dataset

originated from Novak et. al 2006 publication [11]. The measurements were conducted at Syncope and Falls in the Elderly (SAFE) Laboratory at Beth Israel Deaconess Medical Center (BIDMC), Harvard Medical School, Boston, MA, with ethical approval from boards at the Beth Israel Deaconess Medical Center and the Joslin Diabetes Center.

Figure 1: Overview of subject characteristics



From the original dataset of n=86 participants were excluded if there was history of stroke, cardiac disease, significant nephropathy, kidney or liver transplant, renal or congestive heart failure, uncontrolled hypertension, carotid artery stenosis, and neurological or other systemic disorders were excluded.

Information regarding administration of chemotherapy in patient history and consumption of caffeine before testing were not disclosed.

The subjects underwent a 3-day washout of antihypertensive medications in hypertensive T2D participants prior to autonomic testing; prohibition of stimulants/autonomic drugs before testing.

Protocol and hardware

The protocol was performed in a predetermined order. Firstly, subjects rested supine for ten minutes with continuous monitoring of cardiovascular and blood pressure signals before the start of the protocol. The protocol started with deep breathing for one minute. After a short pause, the Valsalva maneuvers were performed consecutively. After Valsalva, a minimum of three minutes rest was taken before the supine to standing test was performed for five minutes. *Figure 3* in results displays a graphical representation of the autonomic testing protocol.

During the protocol, the Electrocardiogram (ECG) and Blood Pressure (BP) were measured. The ECG device used was manufactured by Spacelab (Spacelab Medical, Issaquah, WA). The lead configuration was modified lead II or lead III. The BP was measured by the Finapres finger arterial plethysmograph (Ohmeda Monitoring Systems, Englewood, CO).

Diagnostic labelling

Subjects were classified as CAN or non-CAN by the CART scoring system (see *Table 1: CART testing and CAN scoring*). In this table the criteria for identifying CAN subjects are listed. The outcome of the CART tests was analysed according to Ewing 1985 et al all-abnormal test result is one point and equal or larger than 2 results in CAN classification. In the CART testing the ECG and BP are used during deep breathing, valsalva and supine to standing[12].

Table 1: CART tests and CAN scoring

Test Parameter	Normal = 0	Abnormal = 1
Heart rate (R-R interval) variation during deep breathing	> 1.10	≤ 1.10
Valsalva Ratio (Max to Min HR)	> 1.21	≤ 1.21
Heart Rate Response to Standing (30:15 ratio)	> 1.04	≤ 1.04
Blood Pressure Response to Standing (Fall in systolic BP)	< 20 mmHg	≥ 20 mmHg

There was an omission of sustained hand grip for CAN diagnosis, because this was the only point not tested in the autonomic testing battery in the CART tests. Secondly, in respect to CART criteria, there was a used exclusion 20 mmHg systolic BP drops <20s into the supine to standing. A motivation is the non-neurogenic nature of initial orthostatic hypotension [13].

Signal processing

The time series data extracted from the CART measurements consisted of the following: Deep breathing for one-minute, varying Valsalva maneuvers and a supine to standing test of five minutes.

The labelled epochs of CART time series data had starting and ending markers. To ensure reliability, no marker data was altered. Marker placement was verified by color-coding starting and ending markers of epochs.

To account for mistiming starting/ending marker placement, Valsalva contains 5 seconds starting and ending margin and the supine to standing had an approach with ending window at 240 seconds in case there was no end marker. *Figure 3* in the results section offers a visual insight into the time series data and marker placement for epoch identification.

ECG & HR (V)

The ECG data was sampled ($f_s = 500$ Hz), for filtering the ECG data, a 5th order IIR butterworth filter was used. The filter was setup to exclude phase delay and had drift reducing frequency's for ECG($f_c = 0.5$ Hz , 100Hz) for reducing baseline drift during the protocol [14]. For R-R extraction the Pan- Tompkins algorithm accompanied by an argmax function for syncing optimally with the QRS peak for extraction [15].

In addition to Pan-Tompkins, a Hampel filter has been used to exclude noisy sample outliers. The Hampel filter used was half window of 5 beats, threshold is 3.0 scaled Median Absolute Deviation

(MAD) [16]. This MAD setting was chosen because it was very mild outlier filtering. This enabled the exclusion of accidental grabs of the Pan Tompkins algorithm. For security, only RR intervals within [0.3 2s] were used for analysis, therefore excluding bradycardia < 30 bpm and tachycardia > 180 bpm.

The weighted Savitsky-Golay filter (SGW) length was 7 beats and polyorder of 3, with through Hann window weight distribution ($\cos^4$). This initiated a conservative filtering window to reduce noise related fluctuations from the HR data and enable wrong sample rejection [17]. Together these filtering methods formed the HR data used for analysis of the CART tests.

Feature engineering

Firstly, the features for the CAN scoring in *table 1*, were extracted with the purpose of identifying CAN or non-CAN subjects. On top of these predictions, for practical low effort implementation of the CART test, some features have been extracted from the signal to investigate the prognostic value in ML models. The following paragraph introduces different segments of the protocol; deep breathing, Valsalva and supine to standing and which features have been extracted from these segments of the protocol.

From one-minute-deep breathing, *CAN-scoring for CART testing criterion 1* was extracted, calculating the heart rate (R-R interval) minimum maximum variation (MMHR). Also, the RMSSD and SDNN were

extracted from deep breathing, forming the time domain HRV features. For the HRV frequency domain features, there was an extended analysis. The HR (R-R interval) data has been resampled by cubic spline interpolation and resampling (fresample = 4 Hz) [18]. The following frequency domain HRV features were extracted LF, HF, LF/HF, nLF and nHF were extracted [18].

The valsalva maneuver also yielded some features. The valsalva ratio *table 1: CAN-scoring for CART testing criterion 2* was calculated from the HR data. The maximum HR could be isolated, which was an extractable peak in the signal. After the maximum, a 20 second delay was taken to observe the window for the minimum HR [19]. On top of this a 20 second margin was used for finding the peak around the starting marker for the Valsalva maneuver. The motivation for this approach was to enhance the robustness of marker placement and adhere to valsalva protocol, as can be observed from *figure 3* in the results section.

The supine to standing test also yielded *table 1: CAN scoring criterion 3* the supine to standing test was for analysis. The 15th and 30th R complex from the starting marker in the supine to standing ECG data was taken for 30/15 ratio. Because these require extreme precision and labeling, the exact validity of this approach might be limited. For context, the beat difference between initiation of the markerplacement sometimes was off due to practical delays during the protocol

or human error. Due to the nature of the sub analysis no further steps in unraveling the reliability of the protocol were possible.

table 1 CAN scoring criterion 4 could be derived from the supine to standing test. The blood pressure response to standing has been measured. Because clinical standards currently deem BP changes >20 mmHg systolic in the <20s window of supine to standing as non- neurogenic initial orthostatic hypotension [20,21]. The window from 20-240 seconds was taken for classifying subjects with orthostatic hypotension. This was also separated into 2 windows, to ensure alignment with the current neurophysiological protocols.

Machine learning

The CART scoring system by Ewing was used to classify patients with CAN or non-CAN. There was an abundance of data in the database, patient history, and blood work. Therefore, these features were implemented as part of the ML models. 300+ features were extracted from the database “Cerebral Vasoregulation in Diabetes”. Features with high clinical effort were removed (MRI, CO2 rebreathing, and US doppler). After separation there were 62 features left, consisting of blood draws and patient history.

If features had more than 15% missing data (23 features were thrown out), 34 features left were used for training the ML-models. The data was passed through a

collinearity filter (Pearsons $r > 0.85$), this concluded that the RMSSD or the SDNN were allowed to be in the model due to collinearity.

Because training on 34 features is not according to the current standard of field in statistics. Regarding rule of thumb, multiply parameters by ten to obtain population. The soft variable preselection was done by Elastic net itself, to prevent overfitting hyperparameters and reducing tree depth for RF models to prevent overfitting on artificial noise.

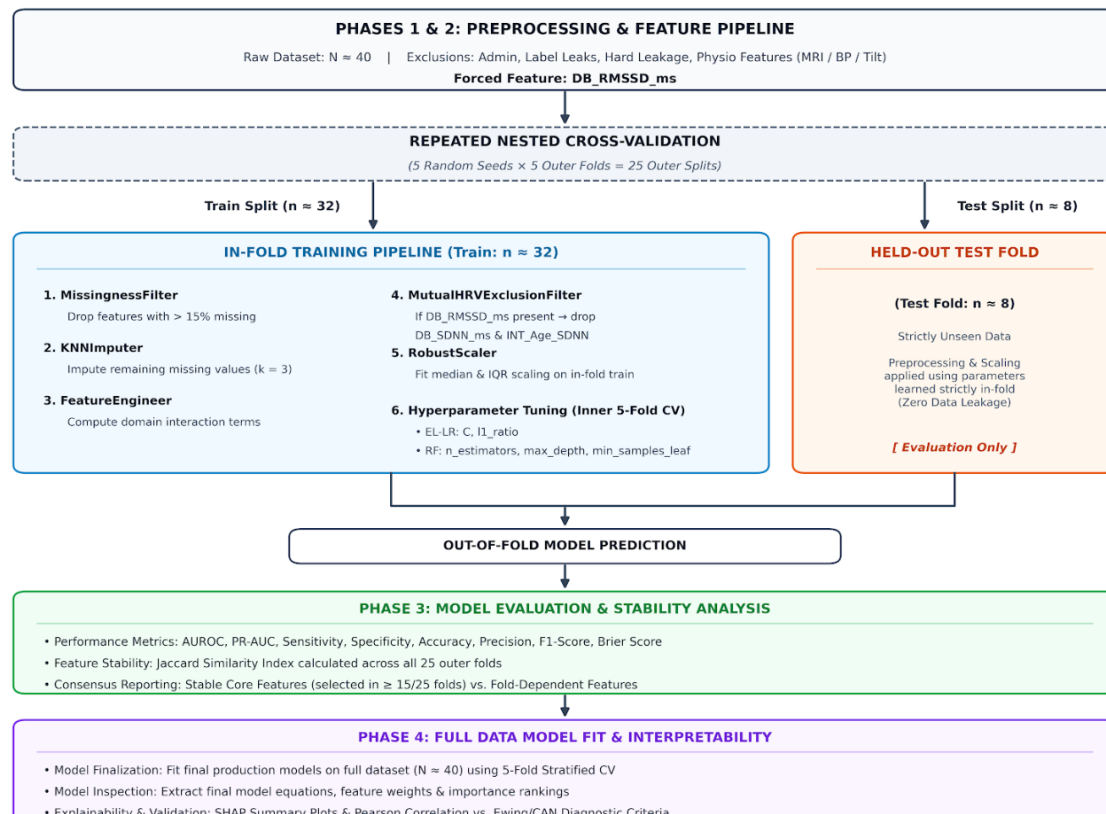
The remaining missing values were filled in by KNN = 3 imputation (inside the cross-validation fold), it fetches three similar rows and is better at maintaining normal data distribution than median imputation [22].

For measuring the performance of the models the AUROC, PR-AUC, Sensitivity, Specificity, F1, have been used as outcome measures. The Jaccard index was used to see if there's significant specificity for CAN in the model. The following results of the Jaccard are vulnerable for FP and FN, therefore the specificity. Information scores per variable can also be used to observe the amount of the result being explained by a certain variable. The concluding result will be displayed by the calibration plots. These following plots are imputed, because of the transparency for the classification results.

To prevent data leakage and ensure robust evaluation, model development utilized a repeated nested cross-validation schematic comprising outer stratified 5-fold splits across different random seeds (yielding 25 test evaluations). Within each outer fold, an inner stratified 5-fold cross-validation optimized hyperparameters via grid search using the area under the receiver operating characteristic curve (AUROC) as the objective metric. The candidate models—Elastic Net Logistic Regression (EN-LR) and Random Forest (RF)—incorporated physiologically linked interaction terms (e.g., vascular toxicity, representing the interaction between hypertension and duration of diabetes mellitus) and applied balanced class

weighting to account for class imbalance. For EN-LR, hyperparameter tuning evaluated regularization strengths ($C \in \{0.05, 0.1, 1.0, 10.0\}$) and mixing parameters ($l1_ratio \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$), whereas RF explored tree depths ($max_depth \in \{3, 5, None\}$), ensemble sizes ($n_estimators \in \{50, 100, 200\}$), and leaf sizes ($min_samples_leaf \in \{1, 2, 5\}$). Finally, the optimal inner model was fitted with an internal threshold calibrator optimizing Youden's J statistic via out-of-fold predictions before evaluating test performance on the held-out outer fold.

Figure 2: Shows an overview of the entire machine learning pipeline.



Results

Figure 3: A Representation of the time series per subject, indicating the ECG, HR and BP data throughout the protocol. With epochs clearly marked by the color-code. Green = Deep breathing, lila/purple = 1st and 2nd Valsalva, blue = suspine to standing test. The Late OH window has been plotted in case there was no markerdata to mark the end of the epoch, the 240s point was not outside of the measurement window.

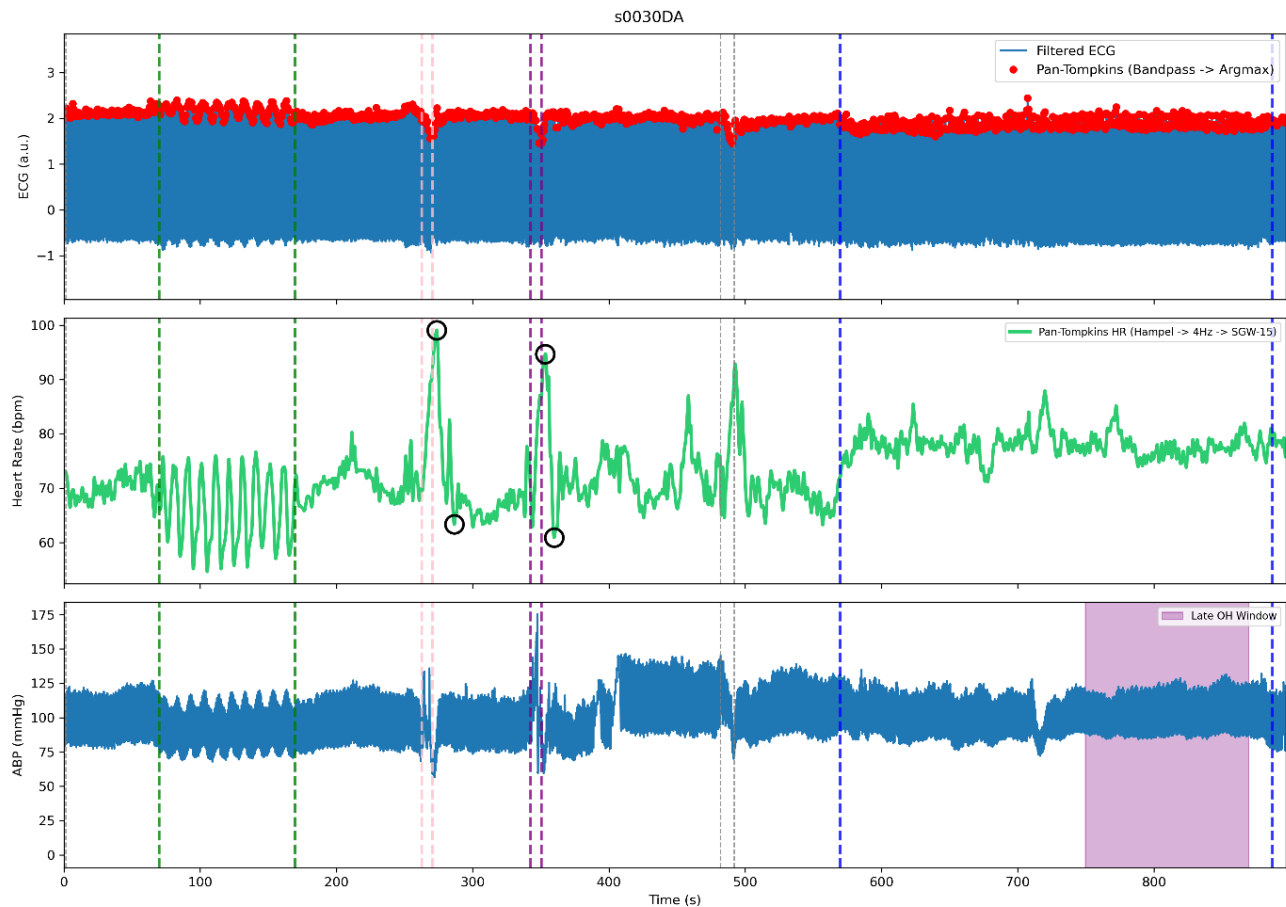


Figure 4 (A + B): Correlation matrix, indicating the pearson correlation coefficient for correlations between model variables and diagnostic variables. With on the Y axis model variables, and X- axis the variables for CART tests implemented for classifying a subject as CAN or non-CAN. This was done for the **Elastic Net Logistic Regression & Random Forest**

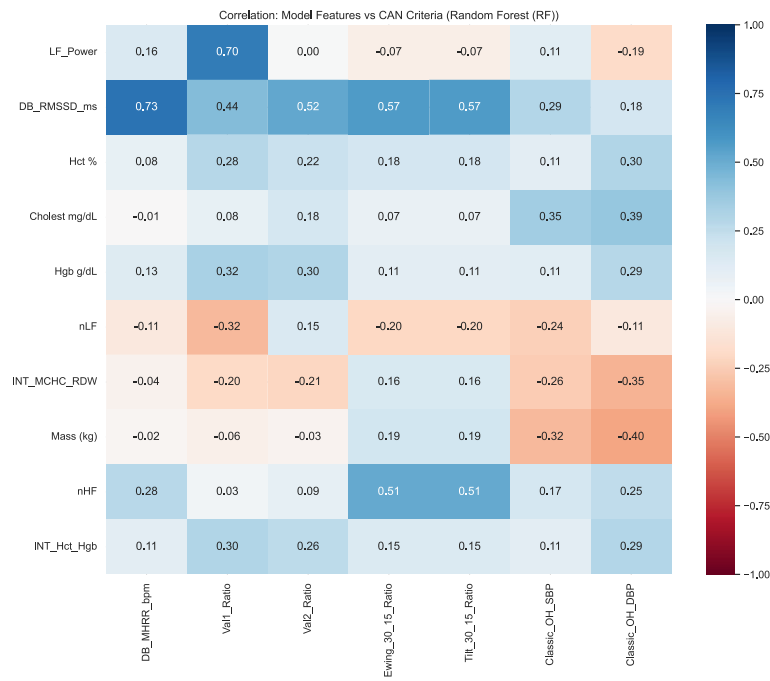


Figure 5 (A+B): SHapley Additive exPlanations (SHAP). This figure shows what the contribution of a certain parameter value on the Y axis is to the model output. The SHAP model enables you to quickly quantify odd results. This was done for the EN-LR & RF models.

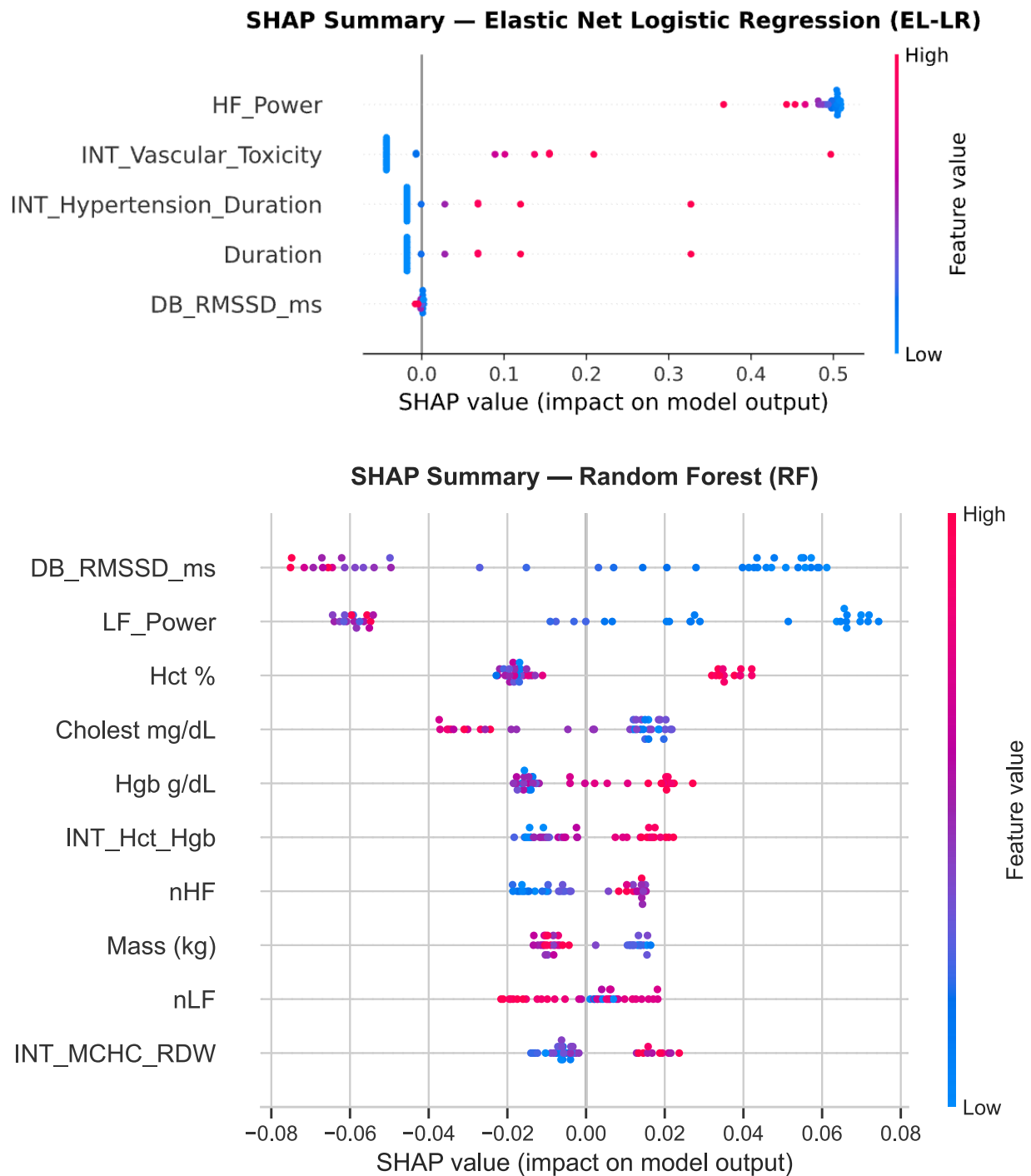


Figure 6: Calibration Curve for RF and EN-LR, This is the TRIPOD AI standard for plotting the logistic regression model on the Y axis the true probability for the model, and on the x-axis the predicted probability. This displays the chance of 1.0 for patients that have the label CAN to also be classified by the model, respectively a zero means a zero percent chance for having CAN according to the model and the labelling. An optimal model follows the diagonal.

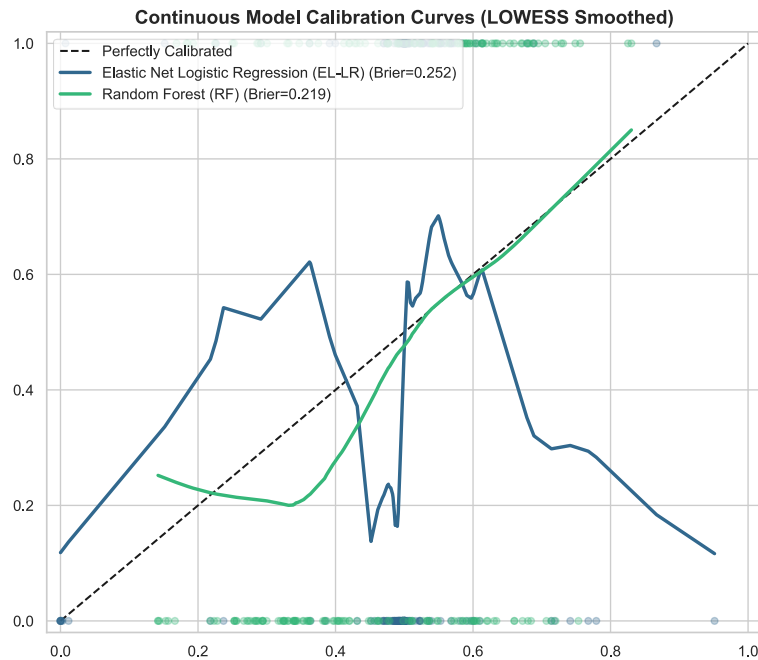
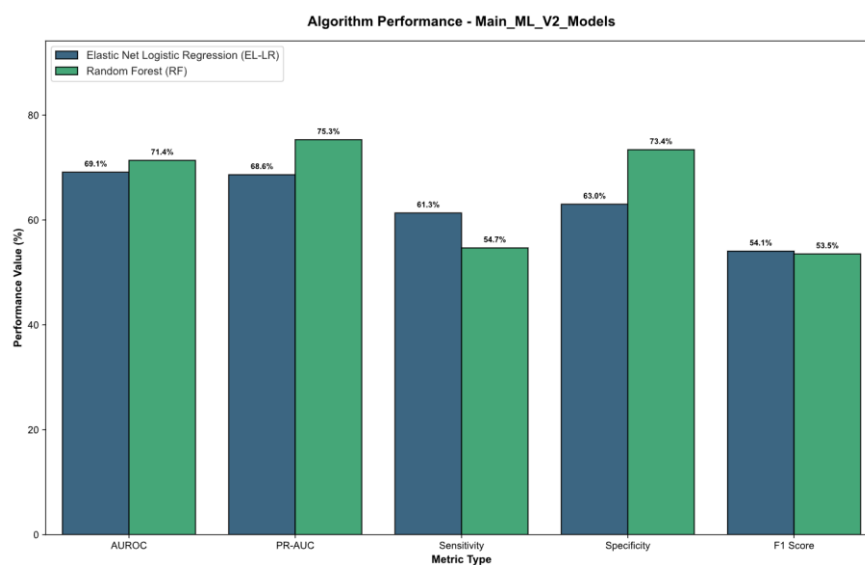


Figure 7: Calibration Performance, auROC, PR-AUC, Sens, Spec and F1 These are the model performance metrics all chosen according to the methods, for estimating the lassification strength of the different model performance.



Discussion

This research focused on CAN identification with variables that require low effort in ML-models. Subjects were labelled with CAN and non-CAN after analysis of the autonomic function (CART) tests. The “low effort variables”, these consisted of blood work, patient history and HRV were inserted in ML-models (RF and EN-LR). This resulted in the HRV time domain feature RMSSD being the best variable for prediction in CAN identification, but the model obtained an AU-ROC < 80% and is therefore not clinically applicable. Therefore, the use of low effort ML models is not feasible as CART replacement for CAN diagnosis.

The RMSSD feature explained the largest quantity of uncertainty in the RF model. The purpose is to capture temporal depth of these years of autonomic decay in neurons with these models. The RMSSD & SDNN represent a decline of autonomic nerve degradation when compared to the non-CAN control group of equivalent age, capturing the decline in autonomic function. However, this approach does not yet account for personal HRV baseline differences and individual temporal changes.

Because T2DM is heterogenous, quantifying the predictive factors is a challenge. This heterogeneity can introduce low explainability of individual variables into the model, especially in low sample size research (n = 40). Because of

the intense nested cross validation protocol, the chance of high overfitting is reduced, but the model violates the ten times population for each variable rule in statistics. This is a display of high heterogeneity in ML models, because RF and EN-LR regardless of the preselection try to include more variables. The variable vascular toxicity. Therefore, a population of (n=40) complicates the training of ML-models in this cross-sectional sub analysis to distinguish the right relationships due to heterogeneity of T2DM.

The current neurophysiological protocols for blood pressure in patients for the identification of Orthostatic Hypotension (OH) are changed. The Ewing protocol used the >20mmHg drop for identifying OH as a CAN identification point according to *Table 1*. Neurophysiological protocols for identification of OH deem a >20 mmHg drop within first 20 seconds as non-neurogenic OH. In this cross-sectional study population with ages between 55 and 75 years we assume that between 15% - 22% suffer from age related OH [23].

Limitations

The sample size for the study was restricted. Therefore, it impacts the reliability of the protocol used in this research. The heterogeneity of CAN in T2DM cannot be properly addressed through machine learning because of the small sample size.

For the current protocol for the CAN identification and labelling in the

autonomic protocol (Ewing 1985 CART battery), OH has been considered as CART requirement. OH can also be associated with age, pointing towards confounding. Research confirms that prevalence of OH is 5-30% in this cross sectional sub analysis [23]. The resulting effects on the CART measurements can be verified by research investigating the impact of the variable OH in CART testing [24]. This research shows that in BP measurements, a drop in mmHg systolic has AU-ROC of 0.589 (0.425- 0.682 95% CI). This yields the same predictive value as a coinflip for CAN identification. Therefore, integrating OH measurements might only introduce noise/randomness into the result of the CART tests.

The original research was documented quite well but had some mistakes. Marker labels were wrongly placed or simply missed, and some participants that must have also be excluded from the original study weren't taken out of the dataset. Because of the lower sample size in the original study compared to the Physio net population, the exact treatment for missing data of the research population remains to be questioned. The inadequate documentation initiates a risk for misinterpreting information or unnecessary exclusion of subjects.

Clinical Implications

For clinical use, HRV time & frequency domain features have useful data separation in identifying CAN. They play a significant role in identifying decline in

autonomic function, in T2DM the RMSSD explained roughly 30% of the mutual information score of the signal. When adding other highly prognostic variables with more research, we'll be able to build a explainable and robust model that'll perform well in either T1DM or T2DM.

In T1DM the signal is expected to be less distorted, the purpose of the ML model was to identify T2DM factors that could be extracted from Electronic Patient Dossier (EPD). Therefore, enabling the CAN risk score before every consultation. In practice we need a high sample size because of the interactions between variables in T2DM to accurately select and identify patterns in the data.

Recommendation

The optimal scenario for clinical context would be highly predictive variables that can be yielded without a large energy investment. Examples are blood tests and patient history due to the volume of T2DM patients that need to be screened for CAN identification.

The data quality needs to be up to standards. This research was done on population T2DM patients. This increases the confounders and effect modifiers making it harder to give a concluding remark on the aspects of disease-causing variables.

Another recommendation is to screen for neuropathy/autonomic neuropathy causing factors. In T2DM alcohol abuse or the incidence of chemotherapy is an

option. If we don't screen for these, we will miss certain links, this expansion is recommended due to low model Jaccard scores and low Entropy for variables outside of SDNN and RMSSD.

Cross sectional research is not the best study design. Doing a cohort study for a couple of years to really capture this autonomic decline in some diabetes patients will contribute greatly. For T1DM looking at HB1Ac every 3 months max and depth in years and time related HRV features and some blood markers like RDW% and WBC. will offer a significant predictor. For T2DM an extension on the number of years (true * years for depth) of hypertension, smoking, appearance of retinopathy, neuropathy, chemotherapy in past. Increasing the sample size and maintaining a good robust record method for increasing data quality is recommended.

Conclusion

The CART tests can't be replaced by low effort variables as substitutes in a clinically generalizable ML-model. The RMSSD HRV feature was a variable that explained most of the signal and was used in every fold, unfortunately the results from the ML-models were not clinically feasible.

AI statement

Antigravity for python coding & consensus for literature search. Further AI has been

used to safeguard and provide a different perspective of literature searches and yielded different sources or results. AI has also been used for creating the ML-methodology overview in the method section.

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